Final Testing and Acceptance Report

Expanded recommendation evaluation and complex workflow results

Group 42 Evidence cutoff 24 October 2026 Revision 2

Overall system acceptance remains deferred. The expanded structured recommendation evaluation completed with the observed functional checks passing. The retained complex workflow experiment still contains 21 final-operation failures. Relevance results remain provisional pending independent human review. [E01-E03]

Area	Latest evidence	Disposition
Basic recommendation scope	40 scenarios: 20 retained and 20 new. There are 36 ranking cases and 4 controls. All observed native final statuses match their registered expectations.	Observed checks passed
Recommendation relevance	K=5: Precision 90.56%, pool Recall 37.33%, NDCG 0.9830. K=10: 85.00%, 59.32%, 0.9541.	Preliminary judgments
Repeated selection	S27 and S29 each have five real-model runs. Graded quality is stable in these two cases while returned identities vary.	Limited stability evidence
Complex workflow	Prior 60-case replay: 39/60 final operations completed, 34/60 every-turn completion, 21 final failures.	Acceptance deferred
Requirement timeout patch	Final intake timeout cases decreased from 33 to 6 in the paired complex rerun. The patch remains local.	Partially mitigated

What this report establishes

The report brings together the expanded Precision, Recall and NDCG results, registered repeat trials, source-fact checks and the unresolved complex-workflow defects. A completed test report does not constitute project-team sign-off or a production release. There is no combined pass rate across the 40 basic and 60 complex scenarios.

Version and publication

The basic recommendation results use unmodified main 4d687e3 and the same real-listing snapshot. The complex suite uses that base plus the registered local timeout patch. The new extension ran only the added scenarios and registered repeats; the retained 20 basic and 60 complex results were reused without new paid calls. This report package and the timeout patch have not been pushed or merged.

Evaluation scope and frozen evidence

Suite	Build and data	Recorded execution
Retained basic cohort	Main 4d687e3. Original 577 judgments plus 42 authorized additions, unchanged.	20 scenarios, 22 native requests including two setup requests; actual paid-model results retained.
Added basic cohort	Same unmodified main and 9 October snapshot. Twenty new needs and a source-only rubric frozen before outputs.	20 main scenarios, four setup requests and eight additional repeat runs: 32 new native requests.
Complex workflow	Same base plus local requirement-timeout patch. Sixty frozen scenarios and 116 authored messages.	Prior HTTP/SSE live replay through intake, recommendation, explanation and knowledge where reached.

What the data counts mean

The database contains 21,431 RENT records from 9 October 2026. Forty basic scenarios represent forty authored user needs. The common reviewed pools contain 1,382 scenario-listing judgments: 619 retained records, 600 new pre-run pairs and 163 identities added from the new saved responses and repeats. There are 1,274 numeric grades and 108 U labels. The 338 main returned positions and 80 additional repeat positions all have numeric grades.

For the new needs, the same transparent rubric was also applied to all 21,431 source records before execution, creating 428,620 frozen rule applications. These are twenty reuses of the source corpus, not 428,620 users, independently authored cases or independent human annotations. Thirty source-only review candidates per new scenario were frozen in advance; sparse qualifying inventories were included in full.

Grading and independence

The new grades use only the authored need and original price, reported bedrooms, advertised area, district and property type. They do not use production scores, output ranks or explanations. Identities returned outside the 600-pair review pools retain their pre-existing corpus grades; only common-pool membership is extended. The original-600 result view is preserved separately.

The extension is schema-aware and grounded in available source fields. It is not a new blind set. Original judgments and the new ordinal rubric use different documented construction methods. Both remain provisional, with independent human review pending. The retained candidate pools include historical outputs and can favor the older selection.

Execution boundaries

The basic suite supplies canonical structured requirements to the unchanged native LiveController, scoring, selection and final gate. Multi-turn cases reference prior native state, but do not test natural-language extraction. Read-only SQLite and frozen real OSM adapters are retained. Explanations, browser UAT, live SQL/graph servers and current rental availability are outside this basic suite. Actual API responses support ten results; the complex application display is capped at five.

Combined recommendation metrics

Ranking means cover 36 primary scenarios. S19, S20, S34 and S36 are controls. Precision and NDCG use all 36; pool Recall uses 35 because S03 has no grade-2/3 identity. Repeat trials do not increase the primary-scenario count. Every K is a prefix of the same actual ten-result response for that scenario. [E01, E02]

K	Precision at K	Pool Recall at K	NDCG at K
1	97.22%	12.80%	1.0000
2	94.44%	19.88%	0.9939
3	93.52%	26.97%	0.9916
4	91.67%	32.15%	0.9877
5	90.56%	37.33%	0.9830
6	89.35%	42.27%	0.9749
7	88.10%	46.74%	0.9700
8	87.15%	51.20%	0.9645
9	86.11%	55.35%	0.9589
10	85.00%	59.32%	0.9541

How a hit and each metric are calculated

A binary hit has grade 2 or 3. Precision at K equals relevant hits in the first K positions divided by K. Missing positions receive no hit. Pool Recall divides those hits by all grade-2/3 identities in the scenario's common reviewed pool. A zero Recall denominator is undefined, rather than zero.

For NDCG, grade gains are 0, 1, 3 and 7. DCG sums gain / log2(rank + 1). NDCG divides DCG by the ideal DCG obtained by sorting the same pool's numeric grades from highest to lowest. U labels are excluded from the ideal pool. Missing result positions have zero gain; an unjudged returned item would keep exact Precision and NDCG unavailable.

How to interpret these results

The first five recommendations have higher average binary precision than all ten, while ten results cover more reviewed relevant identities. This comparison does not independently establish the best product display size. No relevance threshold was predeclared as a release gate.

NDCG measures relative ordering within the judged pool. S03 can score 1.0000 even with zero binary hits because its weak grade-1 sequence matches its ideal pool. NDCG 1.0000 therefore does not mean complete user satisfaction. Source checks establish consistency with stored advertisements, not the current truth or availability of a home.

Cohort results and recall denominators

Both cohorts use main 4d687e3 and the same listing snapshot. Their user needs and label-construction methods differ, so this table is a descriptive coverage comparison. A higher aggregate NDCG after expansion is not evidence of a code improvement. [E01, E02]

K	Retained P	Added P	Retained R	Added R	Retained NDCG	Added NDCG
1	94.44%	100.00%	8.71%	16.67%	1.0000	1.0000
2	94.44%	94.44%	17.41%	22.22%	0.9877	1.0000
3	94.44%	92.59%	26.12%	27.77%	0.9832	1.0000
4	93.06%	90.28%	32.86%	31.48%	0.9806	0.9947
5	92.22%	88.89%	39.61%	35.18%	0.9790	0.9870
6	90.74%	87.96%	45.86%	38.88%	0.9682	0.9816
7	88.89%	87.30%	51.14%	42.58%	0.9625	0.9775
8	87.50%	86.81%	56.41%	46.28%	0.9528	0.9761
9	85.80%	86.42%	61.04%	49.99%	0.9409	0.9769
10	84.44%	85.56%	65.66%	53.34%	0.9317	0.9765

Reviewed pools and source-wide coverage

Headline Recall covers the reviewed candidate identities. The common pool includes returned identities from the registered repeat runs; every repeat uses the same denominator. A separate new-cohort diagnostic uses all known relevant identities under the frozen corpus rubric. It is not interchangeable with pool Recall or a measure of all real-world relevant homes.

New cohort only	Pool Recall	Frozen source Recall
K=1	16.67%	13.73%
K=5	35.18%	20.50%
K=10	53.34%	24.13%

The difference reflects the denominator: some new needs have hundreds of source records meeting their main preferences. The 428,620 rule applications make the new-only source denominator reproducible, while unknown facts remain excluded from known relevance. Original-600 uncertainty views, all numerators and denominators remain in the metric JSON.

Judgment sensitivity remains open

The retained S13 judgment for listing 500229237 remains a human-review priority: its NUS distance is about 5.41 km, beyond the cited grade-2 anchors. Reclassifying it from grade 2 to 1 would lower retained Precision at 10 from 84.44% to 83.89%, and combined Precision from 85.00% to 84.72%. No frozen label was changed for this report.

Repeated cold start and sparse inventory

S27 and S29 each have five actual native runs, including the main run and four registered fresh-session repeats. Request IDs were fixed before execution; native select-v1 derived and recorded the random seeds. All ten runs used cold-start exploration. These are repeated selection observations, not a cold-start on/off causal experiment. [E02]

Case	K	Mean P	Mean pool R	Mean NDCG	Unique IDs	Mean Jaccard
S27	5	100.00%	10.87%	1.0000	16	0.1456
S27	10	100.00%	21.74%	1.0000	35	0.1198
S29	5	100.00%	17.24%	1.0000	10	0.5333
S29	10	100.00%	34.48%	1.0000	26	0.2855

Within these two cases, the minimum and maximum of each quality metric equal its mean across all five trials. At K=10, S27 showed 35 distinct identities and S29 showed 26; lower Jaccard overlap indicates changing identities. Stable graded quality in these two needs does not establish stability for all budgets or preference profiles.

Cases that explain lower precision or ordering quality

Case	Count	P at 10	NDCG at 10	Observed interpretation
S21	10	90.00%	0.7913	One principal-preference miss; fewer top-grade homes in the selected order.
S24	10	100.00%	0.7861	All ten are binary hits, but several secondary-preference matches rank below the pool ideal.
S26	10	30.00%	1.0000	Only three source listings meet the main saving preference; all ten pass the hard limits.
S33	3	10.00%	1.0000	Only three mandatory matches exist, and one meets the principal saving preference.
S35	2	10.00%	1.0000	Only two mandatory matches exist, and one meets the principal saving preference.

S33 returns three homes and S35 returns two, exactly matching their known qualifying inventories. The lists contain no nonqualifying padding. At K=10, the combined 36 primary lists contain 22 empty slots. Fixed-K Precision is 85.00%; mean Precision over filled positions is 88.70%. The measures answer different questions. Complete user acceptance also requires requirement preservation and explanation delivery.

New boundary and preference scenarios

The following twelve cases extend basic coverage using available source facts. Rent figures are SGD per month. Complete prompts, canonical conditions, frozen rubric clauses and 30 pre-run candidates per case are in the English workbook and input files. All twelve returned ten results and passed the native final gate. [E02]

Case	User need and preference order	Type
S21	Rent <= $1,500; >=1 reported bed; >=200 sqft. Prefer <=$1,200, then >=350 sqft.	Boundary
S22	Rent <= $4,000; >=2 beds; area >=700 sqft inclusive. Prefer >=900 sqft, then <=$3,500.	Boundary
S23	Exactly 2 beds in D18/D19; <=$4,500. Prefer >=800 sqft, then <=$3,500.	Boundary
S24	Rent $2,500-$3,500 inclusive; >=2 beds in D23/D25. Prefer >=900 sqft, then <=$3,000.	Boundary
S25	HDB flat; >=3 beds; <=$4,000. Prefer D18/D19, then >=1,000 sqft.	Boundary
S26	D05; >=2 beds; <=$3,500. Prefer <=$3,000, then >=750 sqft.	Boundary
S27	<= $4,500; >=2 beds. Prefer D03/D05 first, then >=900 sqft; other districts allowed.	Preference
S28	<= $4,500; >=2 beds. Prefer >=1,000 sqft first, then D03/D05.	Preference
S29	D18/D19/D20/D23; >=2 beds; <=$4,200. Prefer <=$2,800, then >=900 sqft.	Preference
S30	<= $4,800; >=3 beds. Prefer >=1,100 sqft first, then <=$3,600.	Preference
S31	D18/D19/D23/D25; >=2 beds; <=$4,500. Prefer HDB first, then >=900 sqft.	Preference
S32	D09/D10/D11; >=2 beds; <=$6,500. Prefer D10 first, then <=$5,500.	Preference

Source verification

Across the 245 new final and repeat positions, independent checks passed all 677 mandatory-condition comparisons and 1,470 returned-source-field comparisons. Those counts include repeat positions. They do not verify facts beyond the stored price, reported bedrooms, advertised area, district, property type and URL. The retained source verification covers 173 positions and 439 hard comparisons separately.

New sparse and changing need scenarios

Case	User need or follow-up	Observed final result
S33	D05; >=3 beds; <=$3,800. Prefer <=$3,600, then >=900 sqft. Three known hard matches.	Success; 3 returned
S34	D09; >=2 beds; <=$2,200. No known qualifying source record.	No match; 0 returned
S35	D20; exactly 4 beds; >=1,200 sqft; <=$4,500. Two known hard matches.	Success; 2 returned
S36	D10; >=3 beds; <=$2,500. No known qualifying source record.	No match; 0 returned
S37	Reduce budget from $4,500 to $3,500; retain >=2 beds, D18/D19 and both preferences.	Success; 10 returned
S38	Expand D05 to D03/D05; retain >=2 beds, <=$4,000 and both preferences.	Success; 10 returned
S39	Relax area minimum from 1,000 to 700 sqft; retain >=2 beds, D23/D25, <=$4,200 and preferences.	Success; 10 returned
S40	Make D15 mandatory; retain >=2 beds and <=$5,000. Prioritize >=900 sqft, then <=$4,500.	Success; 10 returned

No match controls and short lists

S34 and S36 correctly returned no_match with no recommendation. S33 and S35 returned all three and two known mandatory matches respectively. Missing bedroom or area facts remain unknown; they are not proof that an otherwise unverified advertisement meets a mandatory need.

What the state update checks cover

S37 through S40 each executed an initial native request, then a final canonical requirements update referencing its saved prior state. The saved requests confirm the authored final conditions, the prior-state reference and requirements version 2. These checks passed. Natural-language parsing of the follow-up, dialogue wording and explanation delivery remain outside these native tests.

The new ordinal rubric

Grade 0 means a known mandatory condition fails. U means missing evidence prevents assigning one exact grade. Grade 1 means mandatory conditions pass but a principal preference is not met. Grade 2 means mandatory and principal preferences pass but a secondary preference is not met. Grade 3 means all mandatory, principal and secondary targets pass. A principal preference outweighs secondary bonuses in this rubric, independently of native weighted scores.

The new cases use reported source fields to stay within the available dataset. Advertised floor area and bedroom count may describe the dwelling even when an advertisement is for a room. This evaluation does not establish whole-unit occupancy, current availability, furnishing, lease terms or an actual user's satisfaction. Human review fields are available in the workbook, with the frozen grade retained separately.

Retained difficult and complex workflow results

The retained complex experiment contains 60 cases and 116 exact authored messages. Its baseline and local-repair runs share main 4d687e3 and the source snapshot; the repair run adds the registered timeout patch. No new complex-suite model calls were made for this extension. The following measures count operational completion. [E03]

Operational measure	Baseline	Local repair	Change
Final authored operation completed	22/60 36.67%	39/60 65.00%	+28.33 pp
Every authored operation completed	18/60 30.00%	34/60 56.67%	+26.67 pp
Completed individual turns	57/116 49.14%	83/116 71.55%	+22.41 pp
Cases ending in intake timeout	33	6	27 fewer

What the completed outcomes contain

Final completed outcome	Cases
Recommendation clarification or unsupported criterion	25
Requirement-intake clarification	10
Other completed outcomes without a fresh property list	4

Thirty-five completed final operations stopped at clarification or unsupported criteria. Some clarifications appropriately expose missing evidence, but their semantic quality has not been independently adjudicated. The other four used knowledge or generic intent clarification without a fresh list.

Six final cases displayed fresh nonempty lists, totaling 30 case-advertisement pairs. All six failed at the explanation stage. No final case both displayed a fresh nonempty list and completed its explanation. This observation does not assign zero semantic correctness to every mixed search, knowledge or clarification case.

Paired recovery and latency

Twenty cases changed from failed to completed, nineteen remained completed, three regressed and eighteen remained failed. The net gain was seventeen cases. Regressed IDs were BT013, BT038 and BT044. These transitions do not establish twenty newly successful rental journeys.

Median turn duration increased from 38.79 to 63.89 seconds; p95 increased from 82.80 to 182.47 seconds. Serial duration increased from 84.02 to 150.78 minutes. The larger deadline allows more completions at a latency cost. No latency service target was agreed as an acceptance gate.

Timeout repair and unresolved defects

Requirement timeout status

PARTIALLY MITIGATED AND VERIFIED LOCALLY. Final intake-timeout cases decreased from 33 to 6. The per-call budget increased from 30 to 60 seconds and the shared intake budget from 90 to 180 seconds. One transient retry is shared across generation, audit and correction. Audit retry reuses its draft; mandatory schema, quote, place and semantic checks remain. Authentication, credit and fatal invalid-request failures are not retried.

The full live rerun supersedes the earlier offline report's pending-live note. Offline evidence includes 14 focused regression tests within 342 passed, 22 skipped and zero failed TypeScript tests, both type checks and a 71-file native-manifest check. Fifteen live operations recovered a transient retry; nine completed and six later failed. The patch remains unpushed and unmerged.

Final failure category	Count	Case IDs
Intake timeout	6	BT005 BT016 BT029 BT037 BT054 BT057
Intake semantic audit	3	BT001 BT022 BT042
Intake schema or extraction validation	3	BT013 BT025 BT044
Native HTTP 422 contract failure	1	BT023
Native MODEL_UNAVAILABLE	2	BT038 BT051
Explanation language or style	5	BT007 BT008 BT012 BT018 BT021
Explanation schema or length	1	BT024

What remains unresolved

Requirement preservation and schema handling remain open: examples include the desk in BT001, studio facilities and daylight in BT022, date-window semantics in BT042, duplicate criteria in BT013/BT025 and an unregistered amenity in BT044. Lowering the semantic audit standard would not preserve those missing needs.

BT023 still needs diagnosis of its exact native 422 request and response. BT038 and BT051 retain model-unavailable business errors; provider, network and application causes have not all been separated. The six list-producing explanation failures are BT007, BT008, BT012, BT018, BT021 and BT024. Requested language and paragraph-length rules remain material to complete delivery.

Repair priorities and acceptance conditions

The following are proposed follow-up checks. They are not additional fixes or completed regressions in this extension. No production code was changed to produce these expanded metrics.

Requirement intake and state preservation

```
src/agents/requirement/execution.ts  extractor.ts  contracts.ts  validation.ts
src/domain/workflow.ts
```

Review and integrate the tested deadline patch, retain phase timing and the bounded retry, and replay all six residual timeout cases. Preserve material user needs, quotes, hard/soft status and multi-turn updates. Unsupported evidence should remain visible as unverified rather than being silently dropped. Duplicate and invalid-amenity failures need specific regression coverage.

Native contract and explanation delivery

```
src/adapters/native-recommendation.ts
recommendation/contracts/requirements.py  validation/rules.py  assembly.py
src/agents/explanation/generator.ts  recovery.ts  validation.ts  contracts.ts
src/model.ts
recommendation/agent/controller.py  model_bridge.ts
```

Diagnose BT023 from the recorded payload before changing serialization or contract validation. Replay the six explanation failures with the saved lists: output must match the current question's language, respect the 600-character paragraph limit, preserve selected order and evidence, and pass recovery validation. Retain bounded structured diagnostics for BT038/BT051 and the previous conversation state.

Exploration and relevance validation

```
recommendation/ranking/selection.py  exploration.py
recommendation/settings/recommendation.yaml
```

Investigate the selected-order loss in S21 and S24 and the retained S08 preference trade-off. A same-main cold-start on/off comparison over predefined seeds is needed to isolate policy effects; the two repeated cases here do not provide that causal result. Ask independent reviewers to check principal/secondary trade-offs, source quality and the S13 borderline judgment. Agree relevance and latency thresholds before treating them as release gates.

Acceptance area	Current disposition
Structured native outcomes and stored hard conditions	Observed checks passed for 40 main scenarios and registered repeats
Recommendation relevance and exploration	Preliminary quantitative results; independent human review pending
Complete dialogue to list and explanation	Deferred; remaining complex failures and successful journey replay needed
Browser acceptance and patch release	Pending browser checks, patch integration and project-team sign-off

Evidence usage and reproduction

ID	Material
E01	evidence/Group42_20_Scenario_Live_Main_2026-10-24.zip. Unchanged original 20-case archive with 577 frozen and 42 supplemental judgments, actual responses, traces and prior checks.
E02	inputs/, raw/, metrics/ and reproduction/ in this expanded package. Twenty new frozen scenarios, source facts and grades, 600 initial review pairs, all 32 new native executions, recorded seeds, API audit and calculations.
E03	evidence/Group42_Blind_Evaluation_Timeout_Fixed_2026-10-24.zip. Unchanged 60-case live replay, baseline comparison, all 116 turns, failure tracker, local patch and offline test evidence.
E04	English formula workbook and reports/Independent_Verification_EN.json. All 440 main/repeat scenario-K rows, 90 macro metric values and 428,620 frozen corpus grades independently checked.

Evidence identities

Main commit

`4d687e3b27e016f622dd487909fc99f56096265c`

E01 original ZIP SHA256

`5cfe10eb150023076728d49ed09d068db30617f9cab6f7b16603cdd6280b4c70`

E03 original ZIP SHA256

`75d8f7f973bdd5e96958c0f50431138cd9c86b269b24311c527dea73e5deedf9`

Local timeout patch SHA256

`3314cee3b488fdabdbb352579d51f5ed146a45327131e7583f3b35a5fbb43cc5`

Actual model usage

Run	Actual HTTP requests	Reported inference usage	Missing usage
New extension and repeats	158	USD 0.079645074	0
Retained basic cohort	109	USD 0.053891106	0
Retained complex rerun	634	USD 0.543085554	78

All 158 new requests returned HTTP 200, with no transport errors or missing usage. The requested route was cline-pass/deepseek-v4.1-flash through Cline; responses reported vmc/deepseek-v4p1-flash-contributor-fallbacks. Backend identity was not independently verified. Reported inference usage is not reconciled billing; missing complex-run usage prevents an exact total. No paid relevance judging or full complex rerun was added.

The independent verifier uses only standard Python and saved evidence: python3 reproduction/verify_expanded_metrics.py. It checks both original ZIP hashes, all frozen input hashes, source-rubric grades and metrics without a model or network. Credentials were removed after execution. Source and database hashes remain unchanged. SHA256SUMS.txt covers the delivered files.

Final disposition: ACCEPTANCE DEFERRED for the complete system. This report is complete as an evidence record. No human relevance sign-off, project-team approval or GitHub publication of this extension is recorded.